

MACE: A Multi-core Agentic Co-design Engine for Autonomous OpenPiton Bring-Up

Rana Umar Nadeem, Samrah Mumtaz, and Muhammad Imran
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Abstract—MACE is an agentic AI system that builds and verifies multicore hardware on its own, running on the CHIA framework. Phase 1 adds `chia_openpiton`, a CHIA adapter exposing OpenPiton’s `configure`, `build`, `simulate`, `run`, and `collect` operations, validated with 208 tests. Phase 2 adds the MACE loop: a planner breaks a hardware objective into tasks, agents run them in parallel, and a deterministic integrator gates every change against Verilator simulation before accepting it. A failure-analysis agent diagnoses failures and the planner replans around them.

We tested MACE against two baselines, manual mesh scaling and a one-shot LLM prompt, on 2x2 and 4x4 meshes, with every result gated by Verilator simulation of the RTL. We added PicoRV32 as a third core, resolving several RTL bugs that kept it from passing under Verilator, and traced what kept multi-tile meshes from passing, including a Verilator 5 scheduling bug that makes CVA6’s cache adapter drop invalidations. With those issues addressed, MACE’s loop passes 2x2 and 4x4 meshes of both Ariane and PicoRV32, while the one-shot baseline fails every configuration it attempts. The artifact pins its toolchain in a Nix flake, entered with a single `nix develop`.

1. Introduction

Current agentic coding systems are typically benchmarked on isolated tasks or toy problems [1], [2], which fails to capture the complexity of actual hardware construction. Bringing up a multicore system means navigating a large RTL codebase (OpenPiton [3] spans hundreds of thousands of lines across network-on-chip, cache-coherence, tile, and platform logic), a fragile toolchain (version skew, cross-compilation bugs, checkout-specific issues), and verification that costs minutes to hours per attempt rather than milliseconds.

MACE is designed for this high-friction environment. Given a hardware objective, such as “bring up a 2x2 mesh” or “verify workload W passes,” MACE plans, implements, builds, simulates, diagnoses failures against Verilator [4] output, and iterates. All validation in this paper is gated by Verilator simulation, not by an agent’s self-report.

1.1. Features

MACE contributes:

- `chia_openpiton` (Sec. 3): a CHIA adapter for OpenPiton with a full `configure/build/run/collect` interface, built the same way as CHIA’s `gem5` and `ESP` adapters, so it can be upstreamed directly.
- The MACE loop (Sec. 4): planning, parallel task execution, deterministic integration, and LLM-driven failure triage, tested end to end against full Verilator runs, with cost, time, and task metrics.
- A third core, PicoRV32 (Sec. 5), brought up by fixing three bugs in its upstream OpenPiton integration.
- An interactive CLI and two further capabilities (Sec. 6): Verilator code coverage as a second verification axis, and a second working LLM backend.

2. Background

CHIA [5] provides the agent, tool, and dispatch layer this project builds on: colocated resource-pinned worker nodes, an LLM abstraction layer spanning multiple backends, and Ray-based [6] remote dispatch and caching. OpenPiton [3] is Princeton’s open-source, silicon-proven manycore research framework; Figure 2 shows a mesh and one of its tiles. Its three tile cores, the latter two added through BYOC [7], are OpenSPARC T1 (`sparc`), Ariane/CVA6 [8] (`ariane`), and PicoRV32 [9] (`pico`). Before this project, CHIA had no OpenPiton adapter.

Related work. LLM-based hardware design has mostly been evaluated at module scale. VerilogEval [2] scores single-module Verilog generation, ChipNeMo [10] adapts LLMs to chip-design tasks, and AutoChip [11], RTL-Fixer [12], VerilogCoder [13], and MAGE [14] feed compiler or simulation results back to one or more agents to repair generated RTL. In software, SWE-bench [15] moved evaluation from isolated functions [1] to repository-level issues. MACE applies the same agent-with-feedback idea at system scale: rather than writing one module against a small testbench, it plans and verifies bring-up of an existing manycore, gated by whole-system Verilator simulation.

3. Phase 1: The OpenPiton Adapter

Design. Figure 3 shows where the adapter sits. `OpenPitonWorkspaceNode` is a CHIA `ColocatedNode` whose resource-pinned remote functions `configure`, `build`, `run`, and `collect` results. OpenPiton writes generated `.tmp.v`

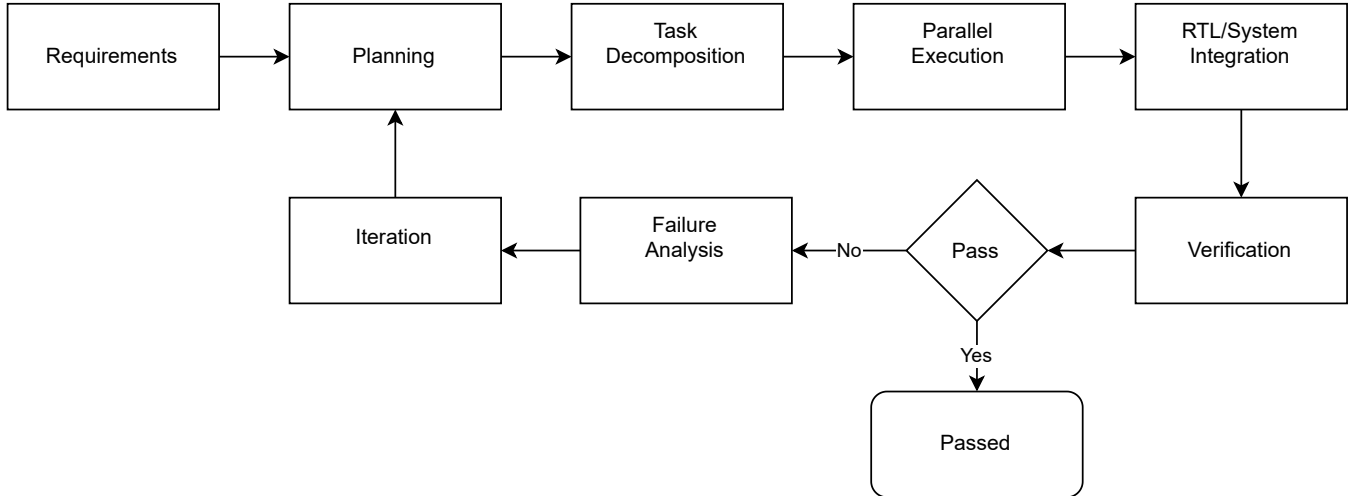


Figure 1. The MACE pipeline. Verification gates every integrated change against Verilator simulation. The run ends when every task passes its gate workload; any failure triggers failure analysis and a replan.

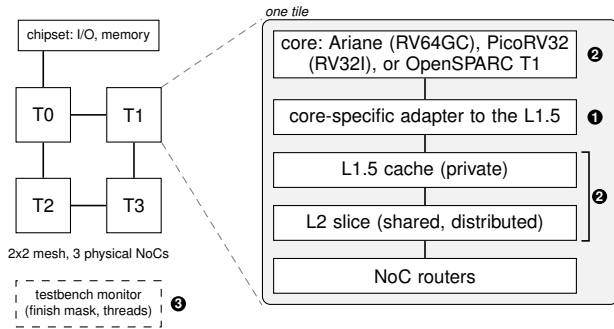


Figure 2. An OpenPiton mesh and one of its tiles. Each core reaches the tile’s L1.5 through its own adapter. The marks show the issues fixed in this work: ❶ CVA6’s L1.5 adapter dropping invalidations under Verilator 5, ❷ PicoRV32’s self-boot gate and the SRAM model’s power-on self-clear, and ❸ the testbench monitor’s 32-bit finish mask and its PicoRV32 thread tracking.

files into its source tree on every build, so each worker gets its own checkout, reserved through one `openpiton` resource unit. Pure parsers read verdicts from OpenPiton’s transcripts, using the strings its RTL monitors emit.

Build-result caching. `build()` skips the simulator when a matching successful-build marker exists: rebuilding an identical configuration dropped from 104s to 0.00s, so an iteration that only changes the test never rebuilds.

Results. 208 tests that need no OpenPiton checkout (pure-unit, and live-Ray with a stub simulator) all pass, and acceptance tests with full Verilator builds pass, including two independent checkouts building in parallel: 176s versus 324s serial. One acceptance test, 2x2 Ariane on a GCP worker, remains blocked by upstream infrastructure dependencies (Sec. 9.1).

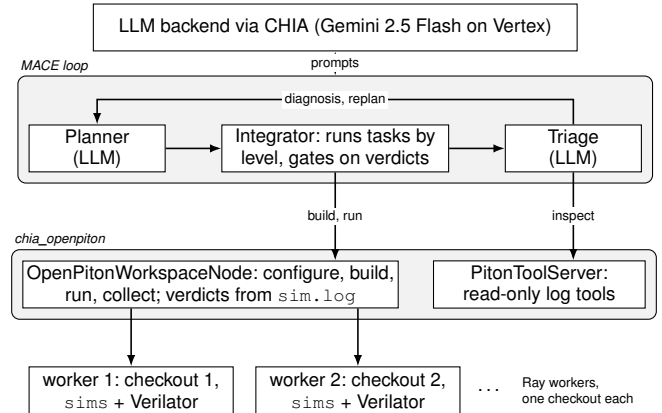


Figure 3. How the pieces fit. The MACE loop drives OpenPiton through `chia_openpiton`, and CHIA and Ray place each checkout’s calls on its own worker.

4. Phase 2: The MACE Loop

Pipeline. Figure 1 shows the loop. A planning agent turns a `MaceSpec` (core, target mesh, gate workloads, objective, budget) into a task DAG, and workers dispatch ready tasks in parallel through Ray, each pinned to its own checkout (Figure 3). A deterministic integrator, with no LLM, runs the tasks in dependency order: each builds its own configuration, including any cache or RTL-define overrides the planner gave it, and runs the gate workload, whose *verdict*, parsed from simulator output, must read *pass*. On any failure, a failure-analysis agent with read-only tools (log grep, file collection, a fixture diff, and an `objdump` symbol check) diagnoses the transcript, and the planner replans around its proposed fix.

Gate workloads surface a memory-visibility bug. Writing the gate workloads (barrier-and-atomic-counter, producer/consumer, hart-id scatter/gather) showed that a

plain volatile read of a shared value was not reliably visible from the same hart shortly after the write, while an atomic fetch-and-add-zero read always was. Every gate workload now reads shared values through this `atomic_read()` helper.

Resource enforcement. MACE checks all three Budget dimensions (iterations, USD, wall-clock) before each iteration starts, and verifies gate-workload checksums before any LLM call or checkout access, so a task cannot tamper with its own grading.

Deterministic replay. MACE tags each task’s LLM, build, and run call, so CHIA’s cache can record and re-serve those decisions at no token or compute cost once caching is enabled.

5. A Third Core: PicoRV32

Each core reaches its tile’s L1.5 through its own adapter (Figure 2), and a core with its own coherent cache, as Ariane has, needs new coherence-adapter RTL. PicoRV32 has no such cache and already has an upstream transducer, which OpenPiton’s CI builds under Verilator but never runs.

Our first pico runs built but never reached a trap. Tracing them found three RTL/testbench bugs (❶, ❷): the core’s self-boot gate, the monitor’s tracking of pico’s tile, and an SRAM power-on self-clear that discarded pico’s first memory writes. With all three fixed, PicoRV32 **passes** under Verilator on 2x2 and 4x4 meshes, every tile reaching `Hit Good trap` on `addi.S`. Its OpenPiton integration has no C compiler, so it runs assembly workloads only.

6. Additional Capabilities

MACE features a standard EDA-style interactive shell (modeled on Yosys/OpenROAD) that validates core compatibility up front to prevent invalid integration attempts. To provide a second verification axis beyond pass/fail, we patched the OpenPiton testbench to call Verilator’s coverage-write API, yielding metrics such as 35.00% line coverage for a 2x2 Ariane run. The agentic loop is also backend-agnostic, completing full passing runs unmodified against Google Vertex Gemini and OpenCode.

7. Evaluation

We compare three approaches on 2x2 and 4x4 meshes: (a) manual mesh scaling with a time log, (b) a single LLM prompt for a configuration applied once with no tools, verification, or retry, and (c) the full MACE loop, with every LLM call on Gemini 2.5 Flash (Table 1). Ariane runs `barrier_atomic.c`; pico runs `addi.S`, since its integration has no C toolchain; SPARC does not yet run under our Verilator (Section 9.1). All three approaches get pico’s `CONFIG_DISABLE_BIST_CLEAR` define: (a) sets it by hand, and the objectives for (b) and (c) name it; (b)’s config line accepts RTL defines, and (c)’s planner requests one through a new `CONFIG_RTL`: directive, which adds it on top of the mesh’s defaults.

TABLE 1. VERILATOR VERDICT AND END-TO-END WALL TIME PER APPROACH. ALL RUNS SHARE ONE BUILD CACHE, SO AN APPROACH PAYS FOR A BUILD ONLY WHEN IT PICKS A NEW CONFIGURATION; (A) COUNTS MACHINE TIME ONLY. [†]INVALID CONFIGURATION (L1.5 SIZE 0), REJECTED BEFORE ANY BUILD.

Core	Mesh	(a) Manual	(b) One-shot	(c) MACE
ariane	2x2	pass, 33.9s	fail, 10.4s [†]	pass, 750.1s
	4x4	pass, 271.9s	fail, 3820.9s	pass, 250.9s
pico	2x2	pass, 9.8s	fail, 12.0s [†]	pass, 70.7s
	4x4	pass, 53.9s	fail, 13.4s [†]	pass, 100.6s

(b) fails every configuration. In three of four, `PitonConfig`’s construction-time validation rejected the LLM’s config before any build: it set the L1.5 size to zero, which every OpenPiton tile needs, and (b) has no retry path. On pico it included the named define both times. At ariane 4x4 it chose oversized caches, including a 4 MB L2 per tile: the build took 35 minutes and the simulation timed out with no tile finished.

Mesh scaling to 2x2 and 4x4. Manual scaling (a) first stalled at 2x2 and never reached 4x4, for two reasons. Under Verilator 5, a known scheduling bug [16] makes CVA6’s L1.5 adapter drop cache invalidations (❶ in Figure 2), so the plain-load exit barrier in `syscalls.c` never sees other tiles’ updates; OpenPiton pins a CVA6 that predates the upstream workaround [17]. Porting that workaround alone passes 2x2 with the unmodified barrier (4 of 4 tiles, 24s), while the original adapter times out with one tile done. Separately, the monitor’s 32-bit `finish_mask` (❷) capped verification at 8 tiles. Our runs poll the barrier with atomic reads rather than port the adapter fix, and widen the mask. With both in place, manual runs pass both sizes, and the loop planned and verified both meshes, every tile confirmed, with no changes to its planning or verification logic (Table 1): 2x2 in 750.1s across four tasks, including an L1D-associativity variant the planner chose to verify, and 4x4 in 250.9s on a cached build. On pico it passes in a single task at both sizes.

Recovery. The detect, diagnose, and replan cycle is exercised only at the mechanism level, with a stub that fails once and then passes; no passing run above needed it. Its one live failure, a pico 4x4 build cached before the `finish_mask` fix, went unrecovered: a build ID covers configuration but not source edits, so all three replans reused the stale build while triage blamed the RTL. A clean rebuild passed.

8. Cost

All experiments behind this paper cost \$37 of a \$300 GCP credit allocation, covering simulation compute and LLM inference; development used a \$20/month Claude subscription, outside that budget. A single loop run is cheap. At 2x2, an ariane run costs about \$0.12 (\$0.04 of LLM tokens and \$0.08 of compute) and a pico run about \$0.02, while a sparc run that exhausts its three-iteration budget would cost an estimated \$0.25, mostly for its one-time build. We price compute as `e2-highmem-8` time (\$0.36/hour) over each

run’s wall time, and tokens at Gemini 2.5 Flash’s \$0.30 and \$2.50 per million input and output tokens. MACE’s own cost counter reads zero because CHIA’s Vertex backend reports no price and omits thinking tokens, more than half of the billed output here, so we measured tokens by re-sending representative planner and task prompts.

9. Limitations and Future Directions

9.1. Limitations

OpenSPARC T1 builds but does not run under our Verilator 5.020: the core never receives its power-on wake-up interrupt. OpenPiton’s own CI recipe stalls the same way here, while CI uses Verilator 4, so the cause is environmental; it remains open. Dispatch to a tailnet-relayed GCP worker is also unreliable: exporting the proxy variables `chia up` sets fixed a first hang, but a later attempt hung again with no outbound traffic, pointing at Ray’s scheduling under a proxied topology.

9.2. Future Directions

All three cores already had upstream adapters to the L1.5; a core without one, such as Ibex or Rocket Chip, needs it written first. The next step is a reusable adapter template built from the shape the three share (a per-core RTL directory, a transducer, and a core-select arm in the tile template), with MACE’s loop verifying each revision of the adapter RTL in Verilator as an engineer writes it. A dashboard over the run database would show dispatch, build, and verification live.

10. Conclusion

MACE shows that an agentic loop can plan, build, and verify multicore hardware, with every result gated by RTL simulation. Along the way we fixed RTL and testbench bugs and added a third core, and the loop then passed 2x2 and

4x4 meshes of both Ariane and PicoRV32. The next steps are a reusable adapter template for new cores and a recovery cycle proven on live failures.

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